25900807 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCECT104, MG1MDCECC100 - CREATIVE ROBOTICS

(2024 ADMISION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** Questions

Each Question carries **1** mark

1. Relate the function of the VIN pin on the Arduino Uno. [U] / [CO1]

   a). It is used for PWM control

   b). It allows external power supply input

   c). It sends serial data to the computer

   d). It resets the Arduino board

2. In the following code, what will happen to the LED? pinMode(9, OUTPUT); analogWrite(9, 128); [U] / [CO1]

   a). The LED will turn on at full brightness

   b). The LED will blink

   c). The LED will turn on at half brightness

   d). The LED will turn off

3. What is the correct syntax to initialize pin 9 as an output? [U] / [CO1]

a). pinMode(OUTPUT, 9); b). pinMode(9, OUTPUT);

c). digitalWrite(9, OUTPUT); d). digitalRead(9, OUTPUT);

4. The function that is called repeatedly in an Arduino sketch is [U] / [CO1]

a). Void start() b). Void run()

c). Void loop() d). Void setup()

5. What is the purpose of using digitalWrite() in robotics projects? [U] / [CO1]

a). To control brightness of LED b). To control actuators like motors and LEDs

c). To read sensor data d). To initialize motors

6. Select the correct operating systems supported by the Arduino IDE. [U] / [CO1]

a). Windows, macOS, Linux b). Windows only

c). macOS only d). Linux only

7. What will be the final value of i after this loop? for (int i = 0; i < 5; i++) { // Do something } [U] / [CO1]

a). 5 b). 4

c). 6 d). 0

8. Identify the correct syntax to set pin 3 as an output in Arduino. [U] / [CO1]

a). pinMode(3, INPUT); b). pinMode(3, OUTPUT);

c). pinMode(3, INPUT_PULLUP); d). pinMode(3, ANALOG);

9. Relate how the serial monitor is used in Arduino programming. [U] / [CO1]

a). To check for compilation errors b). To display data sent by the Arduino board to the computer

c). To monitor power consumption of the board d). To control the speed of code execution

10. Compare INPUT and OUTPUT modes in the pinMode() function. [U] / [CO1]

a). INPUT is for reading signals, and OUTPUT is for sending signals b). Both are used for reading signals

c). Both are used for outputting signals d). INPUT is for digital signals, and OUTPUT is for analog signals

11. Which Arduino pins are used for reading analog input values? [U] / [CO1]

a). A5 b). D5

c). VCC d). GND

12. How do you gradually increase the brightness of an LED using Arduino? [U] / [CO1]

a). Use analogWrite() to adjust the duty cycle b). Use digitalWrite() to set pin states repeatedly

c). Change the resistor value dynamically d). Use pinMode() to adjust brightness

13. What makes voltage divider circuits practical for interfacing sensors with microcontrollers? [U] / [CO1]

a). They amplify signals efficiently. b). They can step down voltage levels without complex circuitry.

c). They reduce noise in the circuit. d). They are only useful for digital inputs.

14. Which of the following represents the correct way to turn OFF the built-in LED on pin 13? [U] / [CO1]

a). DigitalWrite(13, OFF); b). DigitalWrite(13, LOW);

c). DigitalWrite(13, 2); d). PinMode(13, LOW);

15. Which of the following will cause a syntax error in a for loop? [U] / [CO1]

a). for (int i = 0; i < 10; i++) b). for (int i = 0; i == 10; i++)

c). for (int i = 0; i <= 10; i++) d). for (int i = 0; i < 10; ++i)

16. How does the FOR loop structure begin? [K] / [CO1]

a). for(int i = 0; i < 10; i++) b). while(count < 10)

c). if (i == 10) d). loop until (i > 10)

17. Which of the following commands will correctly set pin 9 to 50% duty cycle (2.5 V output) using PWM? [U] / [CO1]

a). analogWrite(9, 1023); b). analogWrite(9, 64);

c). analogWrite(9, 128); d). analogWrite(9, 255);

18. What does the semicolon (;) signify in Arduino code? [K] / [CO1]

a). End of a statement b). Start of a loop

c). Separation of code blocks d). Commenting

19. What function is used to introduce a delay in Arduino programming? [K] / [CO1]

a). delay() b). pause()

c). wait() d). hold()

20. Which of the following functions can be called in both setup() and loop()? [U] / [CO1]

a). pinMode() b). digitalWrite()

c). Serial.begin() d). All of the above

21. What is the primary difference between a while loop and a for loop? [U] / [CO1]

a). A while loop runs indefinitely, while a for loop runs for a fixed number of times

b). A while loop requires a condition, while a for loop does not

c). A for loop includes initialization, condition, and iteration, while a while loop only has a condition

d). A while loop runs faster than a for loop

22. In a voltage divider circuit, if the input voltage is 12V and the resistors are equal in value, what is the output voltage? [K] / [CO1]

a). 9V b). 12V

c). 6V d). 3V

23. Indicate which pin can be used as a ground connection on the Arduino. [U] / [CO1]

a). A0 b). Pin 13

c). GND d). VIN

24. Which of the following is the correct way to print text data on the serial monitor in Arduino? [U] / [CO1]

a). Serial.print("Hello"); b). Serial.display("Hello");

c). Serial.show("Hello"); d). Serial.write("Hello");

25. Which Arduino pin mode is used to control an LED? [U] / [CO1]

a). INPUT b). OUTPUT

c). INPUT_PULLUP d). PULLDOWN

26. Write a code snippet to blink an LED connected to pin 8 using digitalWrite() with a delay of 500ms. [U] / [CO1]

a). digitalWrite(8, HIGH); delay(500); digitalWrite(8, LOW); delay(500);

b). analogWrite(8, HIGH); delay(500); analogWrite(8, LOW); delay(500);

c). analogWrite(8, 128); delay(500);

d). digitalWrite(8, LOW); delay(1000);

27. Indicate the number of analog input pins available on the Arduino Uno board. [U] / [CO1]

a). 8

b). 6

c). 10

d). 4

28. What is the role of 'int i = 0' in a FOR loop? [K] / [CO1]

a). Declares and initializes a variable

b). Specifies the loop condition

c). Increments a variable

d). Declares a constant

29. Which motor driver is known for controlling low-power DC motors? [U] / [CO2]

a). LDR

b). MQ2

c). L298N

d). HC SR04

30. Which Arduino pin type is used to send control signals to a servo motor? [U] / [CO2]

a). Digital

b). Analog

c). PWM

d). Power

31. Which Arduino function is used to read the pulse duration from the Echo pin? [U] / [CO2]

a). pulseIn()

b). digitalRead()

c). millis()

d). delay()

32. Which type of pin does a sensor use for digital signals? [U] / [CO2]

a). VCC

b). GND

c). OUTPUT

d). INPUT

33. What type of motor is used in precise control of angle? [U] / [CO2]

a). DC Motor

b). AC Motor

c). Servo Motor d). Stepper Motor

34. Which pin is generally used to read analog values in Arduino? [U] / [CO2]

a). PWM pins b). Analog pins

c). Digital pins d). Ground pins

35. What does an IR flame sensor primarily detect? [U] / [CO2]

a). Heat b). Light

c). Flame d). Smoke

36. Which pin of the ultrasonic sensor is responsible for sending the sound wave? [K] / [CO2]

a). Trig b). Echo

c). VCC d). GND

37. Which gas does the MQ-2 sensor detect? [U] / [CO2]

a). CO2 b). Hydrogen

c). Nitrogen d). Propane

38. What is the time interval between sending and receiving the signal in an ultrasonic sensor called? [K] / [CO2]

a). Echo time b). Response time

c). Signal duration d). Propagation delay

39. What happens to the resistance of an LDR as light intensity increases? [U] / [CO2]

a). Increases b). Decreases

c). Remains the same d). Fluctuates

40. What is the correct syntax to define PinMode for a trigger pin in Arduino? [K] / [CO2]

a). pinMode(triggerPin, OUTPUT); b). pinMode(triggerPin, INPUT);

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[35X1=35]

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